

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Write a program using JavaScript (arrays, functions)

2. AIM

To write and execute a JavaScript program using arrays and functions.

To implement functions for adding, displaying, and managing items in a shopping list.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic concepts of JavaScript arrays and functions.
 - To create and manipulate an array to store multiple shopping items.
 - To implement functions for adding and displaying items.
 - To use array methods such as `push()`, `join()`, and `length`.
 - To use conditional statements to display appropriate messages based on the number of items.
 - To use JavaScript functions to display the current date and continuously updated time.
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4. THEORY

JavaScript is a programming language used to add logic, functionality, and interactivity to webpages. In this application, JavaScript is used to create an array for storing shopping items and functions for adding, displaying, and managing the items.

The program uses JavaScript concepts such as **arrays, functions, array methods, conditional statements, and the Date object**.

- **Arrays** are used to store multiple shopping items.
- **Functions** are used to perform specific tasks such as adding items and displaying the list.
- **push()** is used to add new items to the array.
- **join()** is used to display the array items as a string.
- **length** is used to find the total number of items.
- **Conditional statements** are used to display messages based on the number of items.
- **Date object** is used to display the current date and time.
- **setInterval()** is used to update the displayed time every second.
- **DOM methods** such as **document.getElementById()** are used to access and update HTML elements.

The basic flow of the program is:

User Input → JavaScript Function → Array Processing → Conditional Checking → DOM Update → Displayed Output

5. ALGORITHM / PROCEDURE

1. Create a new HTML file.
2. Add the basic HTML document structure using `<!DOCTYPE html>`, `<html>`, `<head>`, and `<body>`.
3. Add the webpage title and required HTML elements inside the document.
4. Create an input field to accept shopping items from the user.
5. Create **Add Item** and **Show List** buttons to perform the required operations.
6. Declare an empty array **items** to store the shopping items.
7. Create the **addItem()** function to retrieve the entered item and add it to the array using **push()**.
8. Create the **showList()** function to display the stored items using **join()**.
9. Use the **length** property to calculate and display the total number of items.
10. Use **if...else** statements to display appropriate messages based on the number of items.
11. Create the **showDate()** function to display the current date using the **Date** object.
12. Create the **showTime()** function to display the current time.
13. Use **setInterval()** to update the displayed time every second.

14. Use `document.getElementById()` to access and update the required HTML elements.
15. Save the HTML file and open it in a web browser.
16. Enter different shopping items and verify the array operations, functions, date, time, and displayed output.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>JavaScript Arrays and Functions</title>
</head>

<body>
    <h1>My F1 Driver List</h1>

    <p id="date"></p>
    <p id="time"></p>

    <hr>

    <h2>Enter Driver</h2>
    <input type="text" id="itemInput" placeholder="Enter driver
name">
    <button onclick="addItem()">Add Driver</button>
    <button onclick="showList()">Show List</button>

    <hr>

    <h2>Output</h2>
    <p id="list"></p>
    <p id="count"></p>
```

```
<p id="message"></p>

<script>
    let items = [];

    function addItem() {
        const input = document.getElementById("itemInput");
        const item = input.value.trim();

        if (item === "") {
            alert("Please enter a driver!");
            input.focus();
            return;
        }

        items.push(item);
        input.value = "";
        input.focus();
        document.getElementById("message").textContent = "Driver
added to the grid!";
        showList();
    }

    function showList() {
        const list = document.getElementById("list");
        const count = document.getElementById("count");
        const message = document.getElementById("message");

        list.innerHTML = "<b>Drivers:</b> " + (items.length === 0
? "No drivers added yet." : items.join(", "));
        count.innerHTML = "<b>Total Drivers:</b> " +
items.length;

        if (items.length === 0) {
            message.textContent = "The grid is empty.";
        } else if (items.length >= 5) {
            message.textContent = "The grid is getting packed!";
        } else {
            message.textContent = "The grid looks good!";
        }
    }
}
```

```

        function showDate() {
            const today = new Date();
            document.getElementById("date").innerHTML = "<b>Date:</b>"
+ today.toString();
        }

        function showTime() {
            const today = new Date();
            document.getElementById("time").innerHTML = "<b>Time:</b>"
+ today.toLocaleTimeString();
        }

        showDate();
        showTime();
        setInterval(showTime, 1000);
        showList();
    </script>
</body>

</html>

```

7. CODE EXPLANATION

JavaScript / HTML Component	Description
<!DOCTYPE html>	Defines the document as an HTML5 webpage.
<input>	Accepts the driver name entered by the user.
onclick="addItem()"	Calls the <code>addItem()</code> function when the Add Driver button is clicked.
onclick="showList()"	Calls the <code>showList()</code> function when the Show List button is clicked.
let items = []	Creates an empty array to store driver names.

function addItem()	Defines a function for adding a name to the F1 driver list.
items.push(item)	Adds the entered item to the <code>items</code> array.
document.getElementById()	Accesses HTML elements using their ID.
items.join(", ")	Combines all array elements into a comma-separated string for display.
items.length	Returns the total number of items stored in the array.
if...else if...else	Checks the number of items and displays an appropriate message.
function showList()	Displays the F1 driver list, item count, and corresponding message.
function showDate()	Displays the current date on the webpage.
function showTime()	Displays the current time on the webpage.
new Date()	Creates a Date object containing the current date and time.
toDateString()	Converts the date into a readable date format.
toLocaleTimeString()	Displays the current time in a readable local format.
setInterval(showTime, 1000)	Calls <code>showTime()</code> every 1000 milliseconds to continuously update the time.
innerHTML	Updates the content of HTML elements with the generated output.
alert()	Displays a message when the user tries to add an empty item.
<p>	Displays the date, time, F1 driver list, item count, and messages.

8. TEST CASES AND EXECUTION DATA

Test Case	Test Scenario	Input / Action	Expected Result	Actual Result	Status
TC-01	Open webpage	Open HTML file in browser	F1 Driver List webpage should load	Webpage loaded correctly	Pass
TC-02	Add an item	Enter a driver name and click Add Driver	Item should be added successfully	successfully	Pass
TC-03	Add multiple items	Enter multiple driver names	All drivers should be stored in the array	Items stored successfully	Pass
TC-04	Display shopping list	Click Show List	All entered drivers should be displayed	Items displayed correctly	Pass
TC-05	Check item count	Add 3 items and click Show List	Total Items should display 3	Total Items displayed as 3	Pass
TC-06	Empty input	Click Add Item without entering an item	Alert should ask the user to add an item	Alert displayed correctly	Pass
TC-07	Check grid status message	Add 5 or more items and click Show List	Message should indicate grid status (e.g., "The grid looks good!" or "The grid is getting packed!")	Correct message displayed	Pass
TC-08	Check date and time	Open the webpage	Current date and time should be displayed	Date and time displayed correctly	Pass
TC-09	Check live time	Keep the webpage open	Time should update every second	Time updated successfully	Pass

9. OUTPUT

My F1 Driver List

Date: Sun Aug 30 2026

Time: 8:27:19 PM

Enter Driver

Output

Drivers: charles, lewis, max, piastri, alonso, kimi

Total Drivers: 6

The grid is getting packed!

10. RESULT

Thus, a **JavaScript program using arrays and functions** was successfully developed. The program stores and displays shopping items using arrays, performs operations using functions, calculates the total number of items, and displays the current date and continuously updated time. The program was successfully tested using multiple test cases.